

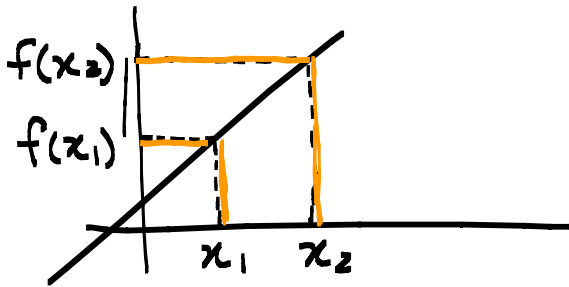
# Unit 4.3 How derivatives affect the shape of the graph. p 290 - 301

Note Title

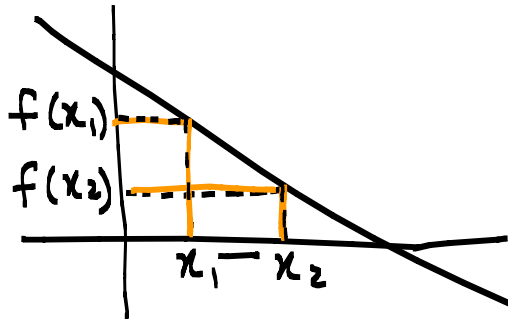
2014/04/07

Remember p 19:

$f$  is increasing on interval  $I$  if  
 $f(x_1) < f(x_2)$  whenever  $x_1 < x_2$  in  $I$ .



$f$  is decreasing on interval  $I$  if  
 $f(x_1) > f(x_2)$  whenever  $x_1 < x_2$  in  $I$ .



Increasing/Decreasing Test p 293

- (a) If  $f'(x) > 0$  on an interval, then  $f$  is increasing on the interval
- (b) If  $f'(x) < 0$  on an interval, then  $f$  is decreasing on the interval.

### Proof (b)

Let  $x_1$  and  $x_2$  be any two numbers in the interval  $I$  with  $x_1 < x_2$ . ( $x_2 - x_1 > 0$ )

$$\text{RTP: } f(x_1) > f(x_2)$$

$$\text{Given: } f'(x) < 0 \text{ on } I$$

Proof: 1.  $f$  is differentiable on  $(x_1, x_2)$   
2.  $f$  is continuous on  $[x_1, x_2]$

By the MVT there exists a number  $c \in (x_1, x_2)$  so that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$\Rightarrow f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

but RHS is negative.

$$\Rightarrow f(x_2) < f(x_1)$$



### Examples:

① Find the interval(s) on which  $f(x) = xe^x$  is decreasing. ( $f'(x) < 0$ )

$$f'(x) = e^x + xe^x = e^x(1+x)$$

$$\text{Where is } f'(x) = \underline{e^x}(1+x) < 0$$

$$e^x > 0 \text{ for all } x \Rightarrow 1+x < 0$$

$$\Rightarrow x < -1$$

$f$  decreases on  $(-\infty, -1)$

(2)  $f(x) = e^{2x} - 6x$  (increasing?)

$$f'(x) = 2e^{2x} - 6$$

$f$  increases if  $2e^{2x} - 6 > 0$

$$\Rightarrow e^{2x} > 3$$

$$\Rightarrow 2x > \ln 3$$

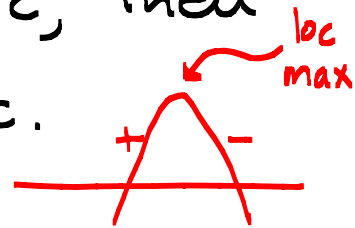
$$\Rightarrow x > \frac{1}{2} \ln 3$$

$f$  increases on  $(\frac{1}{2} \ln 3, \infty)$ .

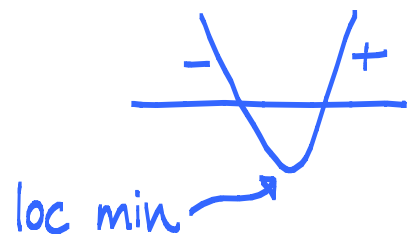
## The First Derivative Test p 291

$c$  is a critical number of continuous function  $f$ :

(a)  $f'$  changes from  $\oplus$  to  $\ominus$  at  $c$ , then  $f$  has a local max at  $x = c$ .



(b)  $f'$  changes from  $\ominus$  to  $\oplus$  at  $c$ , then  $f$  has a local min at  $x = c$ .

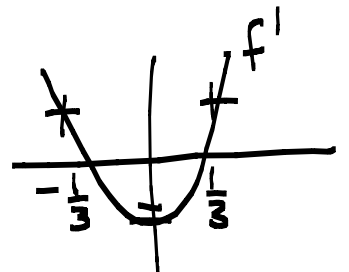


(c)  $f'$  does not change sign at  
then there is no local max/min at  $c$ .

Example:

Find the local extremes of

$$f(x) = 3x^3 - x.$$



CV's:  $f'(x) = 9x^2 - 1 = 0$

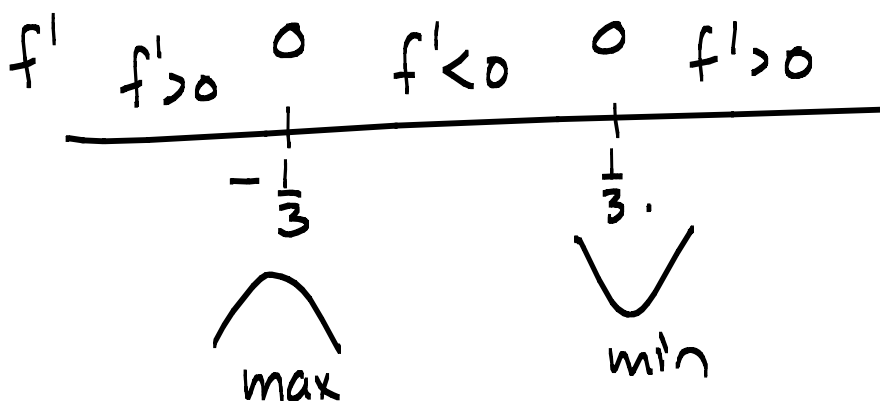
$f'$  is defined everywhere.

$$\text{and } (3x - 1)(3x + 1) = 0$$

$$x = \frac{1}{3}, -\frac{1}{3}$$

So CV's  $\frac{1}{3}$  and  $-\frac{1}{3}$ .

Where if  $f'(x) > 0$  (increasing).



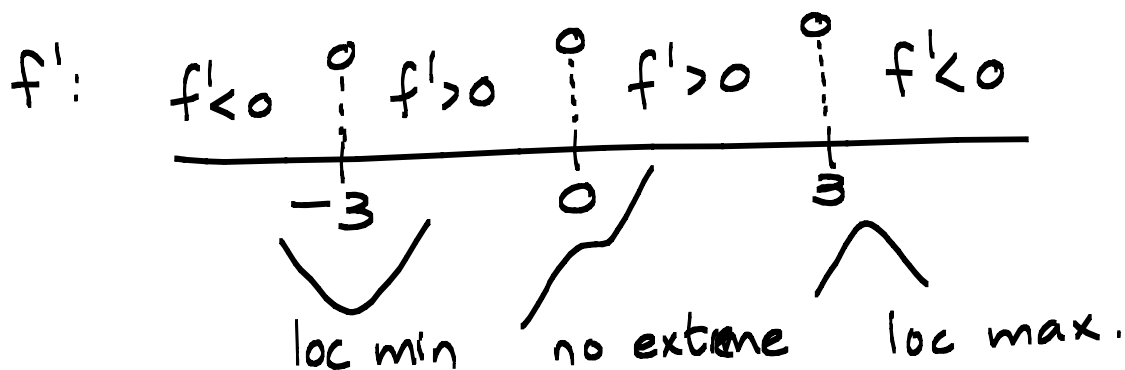
local max at  $x = -\frac{1}{3}$  of  $f(-\frac{1}{3}) = \frac{2}{9}$   
local min at  $x = \frac{1}{3}$  of  $f(\frac{1}{3}) = -\frac{2}{9}$

② local extremes of  $f(x) = -x^5 + 15x^3$

$$\begin{aligned} f'(x) &= -5x^4 + 45x^2 \\ &= -5x^2(x^2 - 9) = \underline{\underline{-5x^2(x-3)(x+3)}} \end{aligned}$$

$f'$  exists on  $\mathbb{R}$  since it is a polynomial.

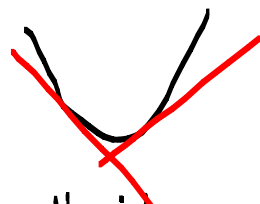
CV's :  $x = 0, -3, 3$ .



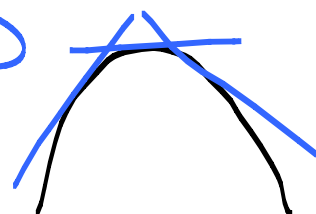
$f(-3) = -162$  is a loc min and  
 $f(3) = 162$  is a loc max.

## Definition p 293

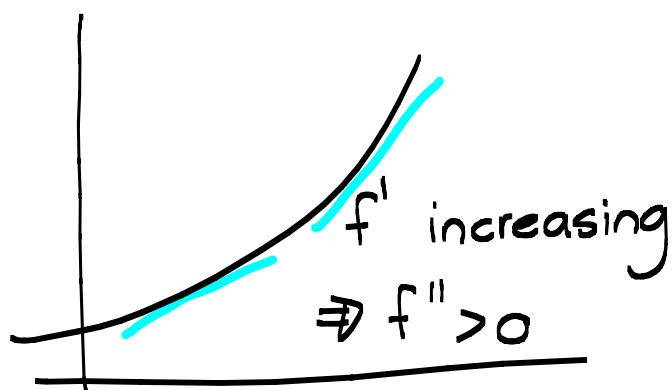
If the graph of  $f$  lies above all its tangent lines: Concave upward on  $I$



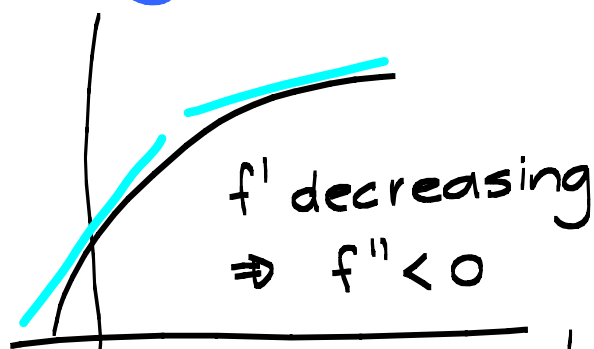
If the graph of  $f$  lies below all its tangent lines: Concave downward



## How do we test concavity?



$f''(x) > 0$    
 $f$  concave upward <sup>local min</sup>



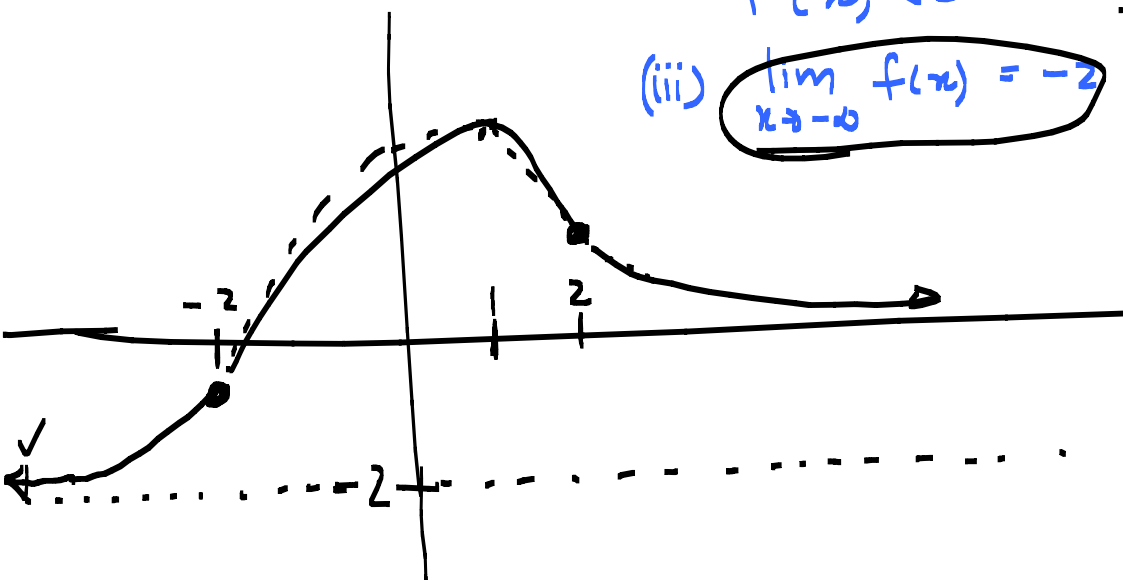
$f''(x) < 0$    
 $f$  concave downward. <sup>loc max</sup>

Sketch a graph satisfying the following:

(i)  $f'(x) > 0$  on  $(-\infty, 1)$   
 $f'(x) < 0$  on  $(1, \infty)$

(ii)  $f''(x) > 0$  on  $(-\infty, -2) \cup (2, \infty)$   
 $f''(x) < 0$  on  $(-2, 2)$

(iii)  $\lim_{x \rightarrow -\infty} f(x) = -2$ ,  $\lim_{x \rightarrow \infty} f(x) = 0$



## The Second Derivative Test p 295

$f''$  is continuous near  $c$ .

(a) If  $f'(c) = 0$  and  $f''(c) > 0$  then  
 $f$  has a local min at  $c$

(b) If  $f'(c) = 0$  and  $f''(c) < 0$  then  
 $f$  has a local max at  $c$ .

If  $f''(c) = 0$  the test fails.

### Examples:

Find the local extremes using the second derivative test.

1.  $f(x) = -x e^{-3x}$

$$f'(x) = e^{-3x}(-1+3x)$$

$f'$  exists every-  
where

$$e^{-3x}(3x-1) = 0$$

$$e^{-3x} \neq 0$$

but  $e^{-3x} \neq 0$  for all  $x$ , so  $3x-1=0$

$$\Rightarrow x = \frac{1}{3}$$



Only CV is  $\frac{1}{3}$ . For 2<sup>nd</sup> der test:

$$f''(x) = 3e^{-3x}(2-3x)$$

So  $f''(\frac{1}{3}) = 3e^{-1} > 0 \Rightarrow f(\frac{1}{3})$  is a local min!



$$2. f(x) = \frac{\ln x}{x} \quad (x > 0)$$

$$f'(x) = \frac{1 - \ln x}{x^2}$$

$f'$  dne if  $x = 0$  but  $x = 0$  is not in the domain of so it is not a CV.

$$\text{and } \frac{1 - \ln x}{x^2} = 0$$

$$\Rightarrow 1 - \ln x = 0 \Rightarrow x = e$$

so CV is  $x = e$ .

$$f''(x) = \frac{-\frac{1}{x}(x^2) - (1 - \ln x)2x}{(x^4)}$$

$$= \frac{-x - 2x + 2x \ln x}{x^4} = \frac{2 \ln x - 3}{x^3}$$

$$\Rightarrow f''(e) = \frac{2 - 3}{e^3} = \frac{-1}{e^3} < 0$$

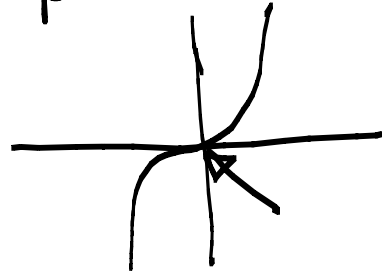
so  $f(e) = \frac{1}{e}$  is a loc max at  $x = e$

## Inflection point p 294

Where the graph changes concavity.

From concave upwards to downwards or  
from concave downwards to upwards.

### Examples:



1. Find the inflection points of

$$f(x) = 1 + 3x + 3x^2 + x^3 \quad \rightarrow \text{look at concavity.}$$

$$f'(x) = 3 + 6x + 3x^2$$

$$f''(x) = \underbrace{6 + 6x} \Rightarrow f''(x) = 0$$

$$f'' \quad \underbrace{f'' < 0}_{\text{blue curve}} \quad \downarrow \quad \underbrace{f'' > 0}_{\text{red curve}} \quad \begin{matrix} 6 + 6x = 0 \\ x = -1 \end{matrix}$$

Changed from concave down to  
concave up  $\Rightarrow$  at  $x = -1$  there is  
an inflection point.

## Examples:

1. The graph of  $f'$  of a function  $f$  is shown.

(a) On what interval(s) is  $f$  increasing or decreasing?

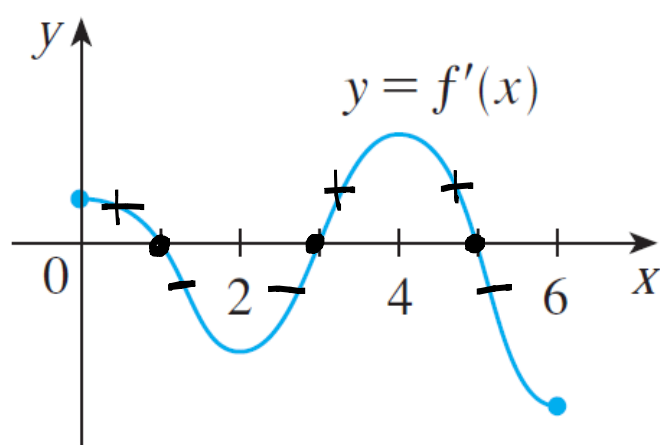
(b) At what values of  $x$  does  $f$  have a local maximum or minimum?

Increase if:

$$f' > 0$$

$f$  increases on  
 $(0, 1) \cup (3, 5)$

$f$  decreases on  
 $(1, 3) \cup (5, 6)$

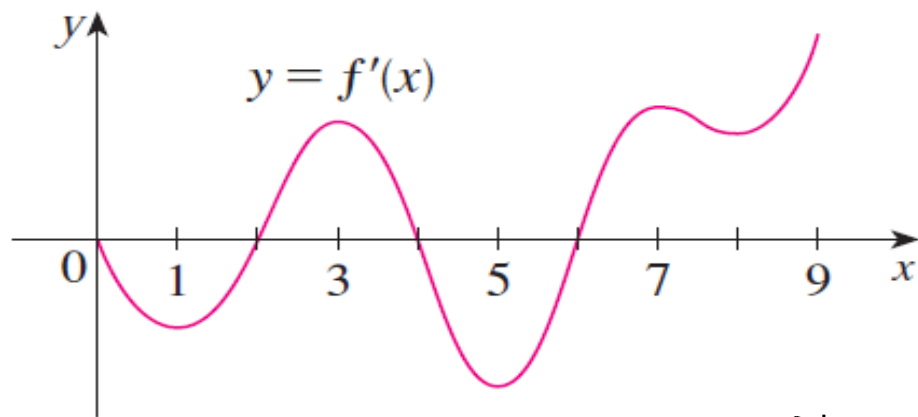


CV's  $f' = 0$

$$x = 1, 3, 5$$

loc max      loc min      loc max

2.



The graph of the derivative  $f'$  is shown.

- On what interval(s) is  $f$  increasing?
- At what value(s) of  $x$  does  $f$  have a local maximum/minimum?
- On what interval(s) is  $f$  concave upwards or concave downwards?
- What are the  $x$ -coordinates of the inflection points of  $f$ .

Sol: (a)  $f$  increasing if  $f' > 0$  so on  $(2, 4)$  and  $(6, 9)$

(b) local max at  $x = 4$   
local min at  $x = 2$  and  $x = 6$

(c)  $f'$  increasing  $\Rightarrow f'' > 0$  (concave up)  
so on  $(1, 3)$ ,  $(5, 7)$  and  $(8, 9)$

$f'$  decreasing  $\Rightarrow f'' < 0$  (concave down)  
so on  $(0, 1)$ ,  $(3, 5)$  and  $(7, 8)$

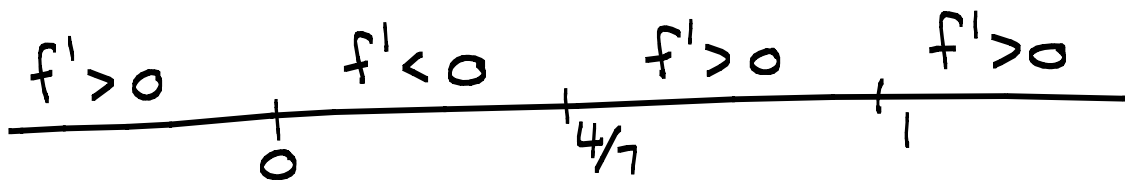
(d) Inflection point where concavity

changes so at  $x=1, 3, 5, 7$  and  $8$ .

3. (a) Find the CV's of  $f(x) = x^4(x-1)^3$   
(b) What does the 2<sup>nd</sup> der. test say about the behaviour of the CV's?  
(c) And the 1<sup>st</sup> der. test?

Sol:

- (a)  $x = 0, 1$  and  $\frac{4}{7}$   
(b)  $f''(0) = 0$   
 $f''(1) = 0$  } Test fails  
 $f''(\frac{4}{7}) > 0 \Rightarrow$  loc min at  $x = \frac{4}{7}$   
(c)  $f'(x) = x^3(x-1)^2(7x-4)$



$f(0)$  is a loc max

$f(\frac{4}{7})$  is a loc min

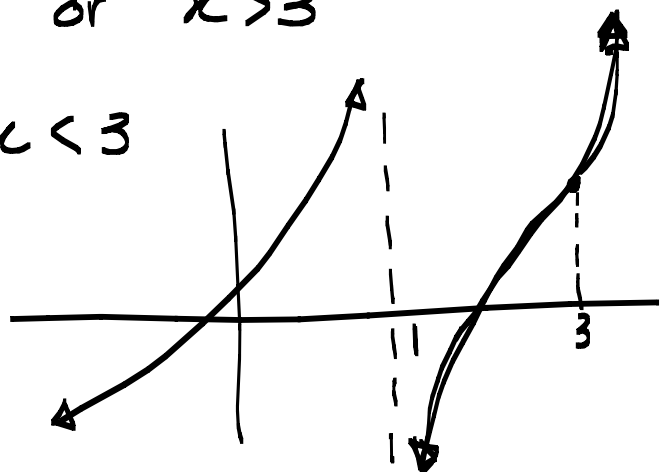
$f(1)$  is neither.

4. Sketch a graph that satisfies:

$f'(x) > 0$  for all  $x \neq 1$

$f''(x) > 0$  if  $x < 1$  or  $x > 3$

$f''(x) < 0$  if  $1 < x < 3$



5. Discuss the curve

$$y = x^4 - 4x^3 + 3$$

with respect to

local extremes, concavity, points of inflection.

Use this to sketch the graph.

1.  $f(x) = x^4 - 4x^3 + 3$

$$f'(x) = 4x^3 - 12x^2 = 0 \Rightarrow x^2(x-3) = 0$$

$f'$  def on  $\mathbb{R}$  so CV's are  $x=0, 3$ .

2. First derivative test

|      |          |              |          |          |         |
|------|----------|--------------|----------|----------|---------|
| $f'$ | $f' < 0$ | $\downarrow$ | $f' < 0$ | $f' > 0$ | loc min |
|      |          | 0            |          | 3        |         |

$f(3) = -24$ .

OR

Second derivative test

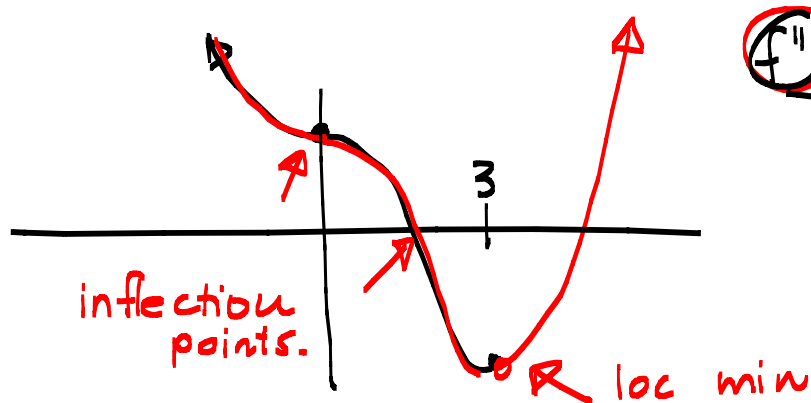
$$f''(x) = 12x^2 - 24x$$

$$f''(0) = 0 \text{ — test fails.}$$

$$f''(3) = 36 > 0 \Rightarrow \text{loc min}$$

Inflection points:

$$f''(x) = 12x^2 - 24x = 0 \Rightarrow x=0 \text{ or } x=2$$



( $f''$ )

|           |           |           |
|-----------|-----------|-----------|
| $f'' > 0$ | $f'' < 0$ | $f'' > 0$ |
|           | 0         | 2         |